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THE JOSEPHSON EFFECT

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Physics 111
June 6, 1967

Introduction

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The most exciting aspect of the Josephson effect experiment is that it allows an undergraduate to duplicate work which was only theory in 1962. Compared to most of the experiments available in the 111 lab, this is much more exciting because it is so modern, and because it allows the student to measure the value of e/h in a macroscopic manner. In order to understand the experiment, it is necessary to have a working knowledge of quantum mechanics. The Josephson effect consists of quantum mechanic electron pair tunneling. These electron pairs behave as bosons, whereas normal single electron tunneling is described by fermi particle characteristics.

It's not all that obvious.

A Josephson junction usually consists of two pieces of superconductor between which is sandwiched a very thin dielectric. The dielectric is conveniently chosen to be the oxide of the superconducting metal. Earlier forms of the junction were made by vacuum deposition of a metal as a strip on a glass slide, allowing it to oxidize, and then vacuum depositing another strip of metal at right angles to the first. The area where the two strips crossed formed the junction. While this method is still used to produce the best junctions, the technique is exceedingly difficult and tedious. The point junction used in this experiment seems much easier to produce and also allows better coupling for the microwaves than the thin film junction. Bridge junctions consisting of two areas of metal film connected by a very narrow strip of film(a few microns wide) have also been used to produce the Josephson

effect.

The Josephson effect can be conveniently divided into three parts: if no potential bias is placed on the junction, a finite current is observed to flow in the junction - this is called the D. C. Josephson effect; if the junction is now biased with a direct current, it produces radio frequency current oscillations called the A. C. Josephson effect; the third is the quantum interference effect in which magnetic flux passing through the area of two or more Josephson junctions in parallel causes the supercurrent to undergo oscillations with quantized changes in the flux.

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Superconduction by Electron Pairs

Metals exhibit resistance to the flow of current due to scattering from vibrating atoms and electrons or from crystaline imperfections and impurities. Superconductors, however, permit free electrons to move in unison through the metal crystal without resistance due to scattering. An electron moving through a superconductor distorts the lattice by attracting positive metal ions in its wake. The concentration of positive charges attracts other electrons so that the net effect is an attraction between electrons. Metals which conduct well at normal temperatures do so because their free electrons have a very weak interaction with the crystal lattice. The weak interaction reduces the scattering of single electrons in normal conductors, but it also reduces the net attraction between electrons which superconductors possess.

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As a result of the interaction, two electrons of equal and opposite spin and momentum are coupled together in weakly bound pairs. They are so weakly bound, that at a few degrees above absolute zero Kelvin, the thermal agitation breaks the bond and destroys superconductivity. Compared to the distance between the lattice ions, the two paired electrons are separated thousands of times farther apart because they are so weakly bound. The Pauli spin principle of quantum mechanics states that two electrons of the same spin can not occupy the same position, and the electron pairs around each ion usually overlap with millions of other pairs. This causes the motions of all the pairs to be correlated. As a result, the center of mass of each pair moves with the same momentum as if all the pairs were locked in unison. They share the same quantum mechanical phase relationship and satisfy the characteristics of bosons.

Phase Relationship in Superconductors

If a voltage V is applied across the ends of a superconducting wire, an electron pair at one end of the wire will have an energy greater by 2eV than a pair at the opposite end. The difference in energy states would produce a phase change throughout the wire and breakdown the electron pair phase bond. This would cause the whole wire to be in a higher energy state. However, if the potential difference across the wire does not exceed a critical point, the wire will keep its lower energy state by conducting current, thereby eliminating the potential across it.

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In the Josephson junction, the phase of electron pairs will be the same in <u>each</u> of the two pieces of superconductor, but the phases of the two superconductors may not be the same. An electron pair which tunnels through the insulator will change phase. If the pair has the same phase upon reaching the other superconductor, it will be successfully accepted there and contribute to a current. The net current will be zero if an electron pair traveling in the opposite direction also matches phase. However, an electron pair which does not match phase after tunneling will be reflected back to the superconductor from whence it came. Thus there will be a supercurrent through the junction when electron pairs succeed in matching phase in one direction, but not in the other. The magnitude of the current has a value of $J_o \sin \delta$, according to Josephson's theory, where δ is the phase difference.

The phase difference can be controlled in the laboratory by adjusting the current that flows through the junction. The phase difference is also directly affected by an external magnetic vector potential. The change of phase in the insulating barrier varies as the magnitude of a magnetic field applied at a right angle in the plane of the barrier.

Since the current is a function of the phase difference and the phase difference is a periodic function of the magnitude of the magnetic field, the current also varies periodically as the magnitude of the magnetic field is changed. Using this fact, John Rowell verified the existence of the D. C. Josephson effect in 1963. He observed that when the magnetic field was slowly increased from zero, the Josephson current went to zero at a value of magnetic field which can be labeled H_o. With a further increase of the flux, the current reached a maxima at $3H_o/2$, $5H_o/2$, etc., and went to zero at $2H_o$, $3H_o$, etc.

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If a current source is attached to the junction and the current is increased above the value of the Josephson current for that particular junction, a potential will be developed across the junction. This potential causes a change in the phase difference across the junction which increases steadily in time. The phase conditions for electron pair transfer through the insulator are constantly changing. First the electron pairs are able to transfer only one direction; the next moment, they are able to transfer only in the opposite direction. This means that supercurrent oscillates back and forth at a very high rate within the junction. The frequency of oscillation is directly proportional to the voltage across the junction. In fact, the frequency is given by $\nu = \frac{2eV}{h}$.

Simple Quantum Mechanical Derivation

The original paper by Josephson in which he analyzed the possibilities of quantum mechanical electron pair tunneling in superconductors is extremely hard for a student to understand. A much clearer explaination is presented in the last chapter of Feynman's Vol. III. The following explaination is derived from this source and from Kittel's notes on Solid State Physics.

D. C. Josephson Effect

Consider a simple Josephson junction where there is a wave function ψ_1 associated with one side of the junction and a function ψ_2 with the other. If the superconductors are identical and at the same potential, the quantum equations for the motion of electrons in the conductors are:

$$i\hbar\dot{\psi}_1 = \hbar T\psi_2 \quad ; \quad i\hbar\dot{\psi}_2 = \hbar T\psi_1$$

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in which the transfer interactions of the electron pairs is represented by $\hbar T$. As the insulating layer becomes thicker than about 10 Å, T goes to zero and ther is no electron pair? tunneling.

Now if we substitute for ψ_1 and ψ_2 the values:

$$\psi_1 = \sqrt{\rho_1}\, e^{i\theta_1} \quad ; \quad \psi_2 = \sqrt{\rho_2}\, e^{i\theta_2}$$

where ρ is the electron density and θ is the phase, then

$$\dot{\psi}_1 = \tfrac{1}{2}\rho_1^{-\frac{1}{2}} e^{i\theta_1}\dot{\rho}_1 + i\psi_1\dot{\theta}_1$$

and

$$i\hbar\dot{\psi}_1 = \tfrac{1}{2} i\hbar\rho_1^{-\frac{1}{2}} e^{i\theta_1}\dot{\rho}_1 - \hbar\psi_1\dot{\theta}_1 = \hbar T\rho_1^{\frac{1}{2}} e^{i\theta_2}$$

Multiply through by $\hbar^{-1}\rho_1^{\frac{1}{2}} e^{-i\theta_1}$ to obtain:

$$\tfrac{1}{2} i\dot{\rho}_1 - \rho_1\dot{\theta} = T(\rho_1\rho_2)^{\frac{1}{2}} e^{i\delta}$$

where $\delta = \theta_2 - \theta_1$

Now if we equate the real and the imaginary parts of the above equation:

$$-\rho_1 \dot{\theta}_1 = T(\rho_1\rho_2)^{1/2}\cos\delta \quad ; \quad \dot{\theta}_1 = -T\left(\frac{\rho_2}{\rho_1}\right)^{1/2}\cos\delta$$

$$\tfrac{1}{2} i\dot{\rho} = T(\rho_1\rho_2)^{1/2}\sin\delta \quad ; \quad \dot{\rho}_1 = 2T(\rho_1\rho_2)^{1/2}\sin\delta$$

Under the conditions we have stated, $\rho_1 = \rho_2 = \rho_0$, so: $\dot{\theta}_1 = \dot{\theta}_2$

and $$\dot{\rho}_2 = -2T(\rho_1\rho_2)^{1/2}\sin\delta$$

Now, the current through the junction must be proportional to $\dot{\rho}_1$ or $-\dot{\rho}_2$ so:

$$J = J_0 \sin\delta$$

where J_0 is dependent on T, a characteristic of the junction. With zero potential, the current through the junction will have a value between J_0 and $-J_0$ depending on the value of δ, the phase difference.

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A. C. Josephson Effect

If the junction is now biased with a D. C. voltage, the equations of motion must be rewritten to give:

$$i\hbar\dot{\psi}_1 = -eV\psi_1 + \hbar T\psi_2 \quad ; \quad i\hbar\dot{\psi}_2 = eV\psi_2 + \hbar T\psi_1$$

And, following the same steps as above:

$$\tfrac{1}{2} i\hbar\rho_1^{-1/2} e^{i\theta_1}\dot{\rho}_1 - \hbar\psi_1\dot{e}_1 = -eV\rho_1^{1/2}e^{i\theta_1} + \hbar T\rho_2^{1/2}e^{i\theta_2}$$

or $$\tfrac{1}{2} i\dot{\rho}_1 - \rho_1\dot{\theta}_1 = -eV\rho_1\hbar^{-1} + T(\rho_1\rho_2)^{1/2}e^{i\delta}$$

then: $$\dot{\rho}_1 = 2T(\rho_1\rho_2)^{1/2}\sin\delta \quad ; \quad \dot{\theta} = \left(\frac{eV}{\hbar}\right) - T\left(\frac{\rho_2}{\rho_1}\right)^{1/2}\cos\delta$$

also,

$$\tfrac{1}{2} i\dot{\rho}_2 - \rho_2\dot{\theta}_2 = eV\rho_2\hbar^{-1} + T(\rho_1\rho_2)^{1/2}e^{-i\delta}$$

$$\dot{\rho} = -2T(\rho_1\rho_2)^{1/2}\sin\delta \quad ; \quad \dot{\theta} = -\left(\frac{eV}{\hbar}\right) - T\left(\frac{\rho_1}{\rho_2}\right)^{1/2}\cos\delta$$

Again, $\rho_1 = \rho_2 = \rho_0$ and using this fact in the values of $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$, we find that:

$$\dot{\delta} = -\frac{2eV}{\hbar}$$

so

$$\delta = \delta_0 - \frac{2eVt}{\hbar}$$

and the current is given by:

$$J = J_0 \sin\left[\delta_0 - \frac{2eVt}{\hbar}\right]$$

oscillating with a frequency ω: $\omega = \frac{2eV}{\hbar}$

This means that everytime an electron pair tunnels through the insulating barrier, a photon of energy $\hbar\omega = 2eV$ must be emitted or absorbed.

By applying both a direct current bias and a small a. c. voltage with a very high frequency, we obtain:

$$V(t) = V_0 + v\cos\omega t$$

which makes $\delta(t)$ thus:

$$\delta(t) = \delta_0 - \frac{2eV_0t}{\hbar} - \frac{2ev}{\hbar\omega}\sin\omega t$$

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then $$J = J_0\left\{\sin\left[\delta_0 - \frac{2eVt}{\hbar}\right] - \frac{2ev}{\hbar\omega}\sin\omega t\cos\left[\delta_0 - \frac{2eV_0t}{\hbar}\right]\right.$$

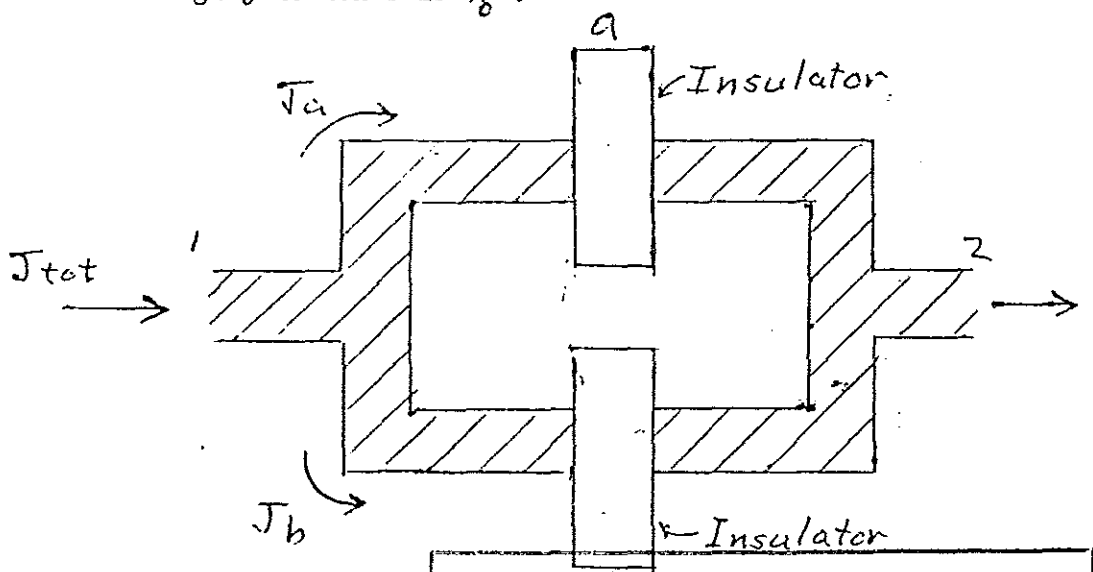
The first term on the right has a time average value of zero, but the second term does not if $\omega = \frac{2eV_0}{\hbar}$ so, there will be a direct current through the junction if a small $(v/V_0 \ll 1)$ rf voltage is induced across it with frequency $2eV_0/\hbar$.

Macroscopic Quantum Interference

The phase difference δ around two Josephson junctions connected in parallel which encompass a magnetic flux $\phi = \oint A \cdot d\ell$ is:

$$\delta = \frac{2e}{\hbar c}\oint A \cdot d\ell$$

The following refers to the schematic diagram of a double junction seen below. The phase difference of current through junction a is δ_a and through junction b is δ_b .



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When flux passes through the interior of the above circuit, it changes the phase relationship of the junctions so that :

$$\delta_b - \delta_a = \frac{2e}{\hbar c}\phi$$

or

$$\delta_b = \delta_o + \frac{e}{\hbar c}\phi \quad \text{and} \quad \delta_a = \delta_o - \frac{e}{\hbar c}\phi$$

$$J_{tot} = J_a + J_b = J_o\left\{\sin\left(\delta_o + \frac{e}{\hbar c}\phi\right) + \sin\left(\delta_o - \frac{e}{\hbar c}\phi\right)\right\} = J_o \sin\delta_o \cos\frac{e\phi}{\hbar c}$$

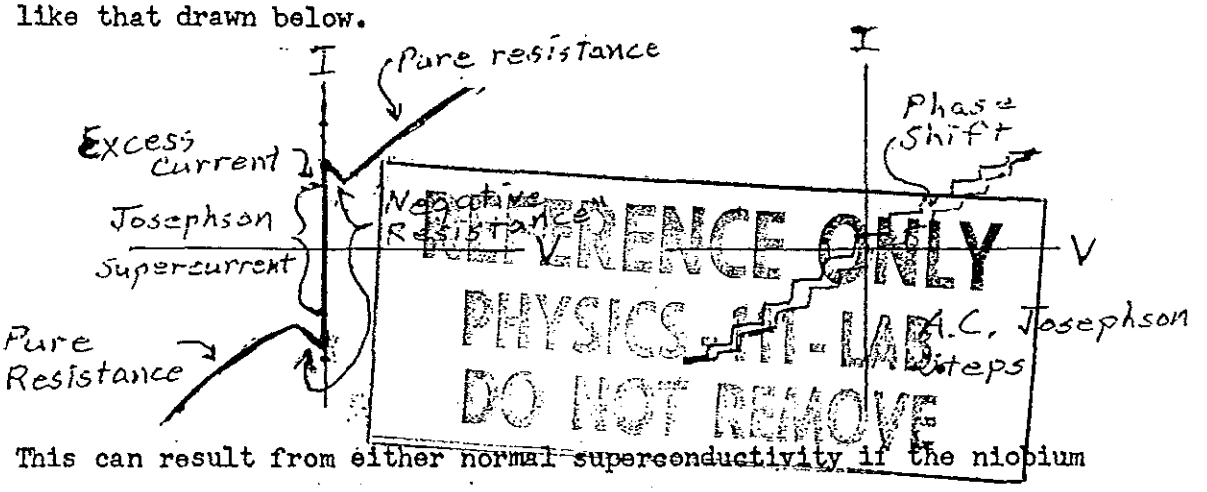
$J(\phi)$ will have maxima at $\frac{e\phi}{\hbar c} = n\pi$ where n = integer

or $$\phi = \frac{n\pi\hbar c}{e} = n\frac{hc}{2e}$$

This effect makes the double Josephson junction useful as a very accurate magnetometer, capable of measuring a field as small as 2×10^{-7} gauss.

Application of Theory to the Experiment

When the point junction is properly adjusted in the absence of external microwaves, it is possible to obtain a current voltage trace like that drawn below.



This can result from either normal superconductivity if the niobium point has pierced the oxide layer or from D. C. Josephson superconductivity. If the trace is due to the D. C. Josephson effect, the vertical slope supercurrent will oscillate in magnitude quite rapidly as a large magnet is moved toward the dewer. The magnet will have no effect on the trace if it has been caused by the niobium wires shorting. In a good point junction, a 2000 gauss magnet will show an observable effect as it is moved at a distance of approximately 10 feet from the dewer. For the D. C. Josephson effect, the magnet causes a phase shift between electron pairs and this changes the current that flows through the junction. Also, due to irregularities in the niobium metal surfaces, there are actually several point junctions at the end of the wire. This means that there will be quantum mechanical interference when magnetic flux passes through the finite areas formed by the sub-points. This was described above for the case of a double point junction, but here we have an effective multiple point junction which produces an equivalent multiple slit interference

pattern in the current as the flux changes.

The pure resistance part of the trace (that part of the trace where there is a definite slope) occurs when the current through the junction exceeds the Josephson supercurrent and a potential appears across the junction. We are using a 60 cycle sine wave to sweep current through the junction, so there are two parts to the resistance curve. It is interesting to note that the vertical current line is often seen to have a value greater than the superconducting current. Then, that part of the trace which goes from the end of the vertical current to the first of the pure resistance indicates a "negative resistance". See figure above.

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The part of the curve which I have called pure resistance is simply where the electron pair phase locking has broken down and single electron tunneling has begun. This trace is also observed in normal superconductivity.

As the microwave power is turned up by adjusting the attenuator, the D. C. Josephson effect breaks up into steps where each of the changes in voltage is equal.(see figure). According to the theory, the size of each voltage step is $V = h\nu/2e$ where ν is the microwave frequency. The explaination for this effect is that the voltage which appears across the junction is actually the summation of two voltages. We are sweeping the junction with a 60 cycle voltage and also with a very small voltage which varies at about 22.3 gigacycles per second. The microwave voltage frequency modulates the Josephson supercurrent since the Josephson frequency is a function of applied voltage. The frequency modulation causes the Josephson current to contain components of many frequencies. When the sum of the Josephson frequency plus or minus the microwave frequency plus or minus the component frequencies plus or minus all the

the integral harmonicsof each of these frequencies equals zero, there will be a direct current flow through the junction. Each time all the frequencies add to zero, a step occurs. In simpler terms, the first step occurs when the microwave frequency or an integral multiple of it equals the Josephson frequency. The amplitude of the nth step decreases according to an equation of the nth order Bessel function $J_n(2ev/h\nu_m)$ where v is the voltage of the microwave. If the microwave attenuator is turned completely off (zero attenuation), the steps will sometimes completely disappear. Occassionally, it was possible to observe a step which was twice as long as the others, indicating that ~~single electron tunneling was occurring.~~ Never

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If the point is very carefully adjusted, it will be possible to observe the A. C. Josephson effect without applying external microwaves. The steps are found on the pure resistance part of the trace and result from the Josephson rf produced by the points beating against its own oscillations inside the resonant cavity formed at the end of the probe. These steps have a very small vertical magnitude (current change) because so little power is radiated from the point.

We were bothered by a phase shift in our cables which caused one half of the 60 cycle sweep to be shifted slightly from the other half so that we obtained a double trace. The steps on each of these traces were consistently the same and the phase shift did not effect the experiment, but was rather annoying. Apparently it was caused by impedance and capacitance i the cables.

APPARATUS AND EXPERIMENTAL PROCEDURE

By

W. H. Putnam

Niobium points

1/2" lengths of niobium wire. File with fine-toothed file in drill press at medium speed. Smooth with emery paper. Points must be etched in NaOH (weak soln) to remove iron filings. Leave points in air for at least a day to oxidize. When a point has been blunted in experiment, do not remove but break off and sharpen again as only a limited supply of niobium wire is available. Best shape is like a phonograph needle; narrow but slightly rounded at the end.

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Apparatus

1. Oscilloscope
2. Low-level preamplifiers
3. K-band microwave oscillator
4. Detector
5. Wave-meter
6. Dewar flask
7. Josephson junction probe
8. Power supply for probe

1. The oscilloscope (Tektronix Type 543B with dual-trace plug-in unit--U.C. code 65-10-07405) is used to make an X-Y plot of current vs. voltage. This particular one has been calibrated to 100.00 .01 volts.

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One of these amps is borrowed from Rad Lab and must be returned. Other one on

2. The preamplifiers (Tektronix Type 122) are wide-band (40 kHz-.2 Hz) differential input amplifiers. In operation one input (pin 1) is grounded to pin 3. The two amplifiers are to provide the necessary .1 mv/cm. sensitivity. Measured amplification factors are 390-3900 horizontal and 142-1420 vertical.

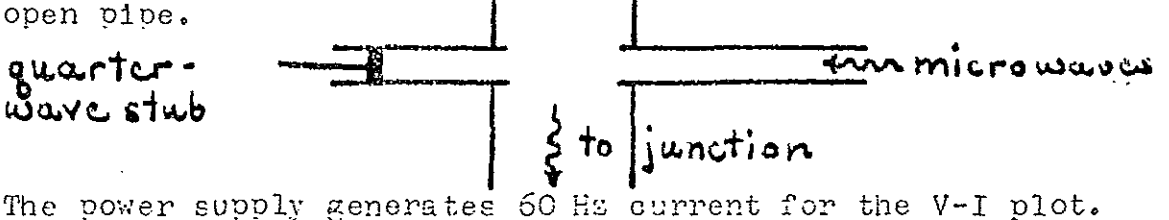
3. Microwave power is supplied by a K-band klystron oscillator (nominal output 20 mw at 22.2 GHz.) The power is fed into a directional coupler (with detector), attenuator, and quarter-wave stub to provide weak coupling of microwaves to the junction at the bottom of the coaxial probe.

4. The crystal detector is an F-R Type K205BF with 1 volt output at maximum klystron power.

5. We have no wave meter at the present time (the former one was outside the range of the klystron).

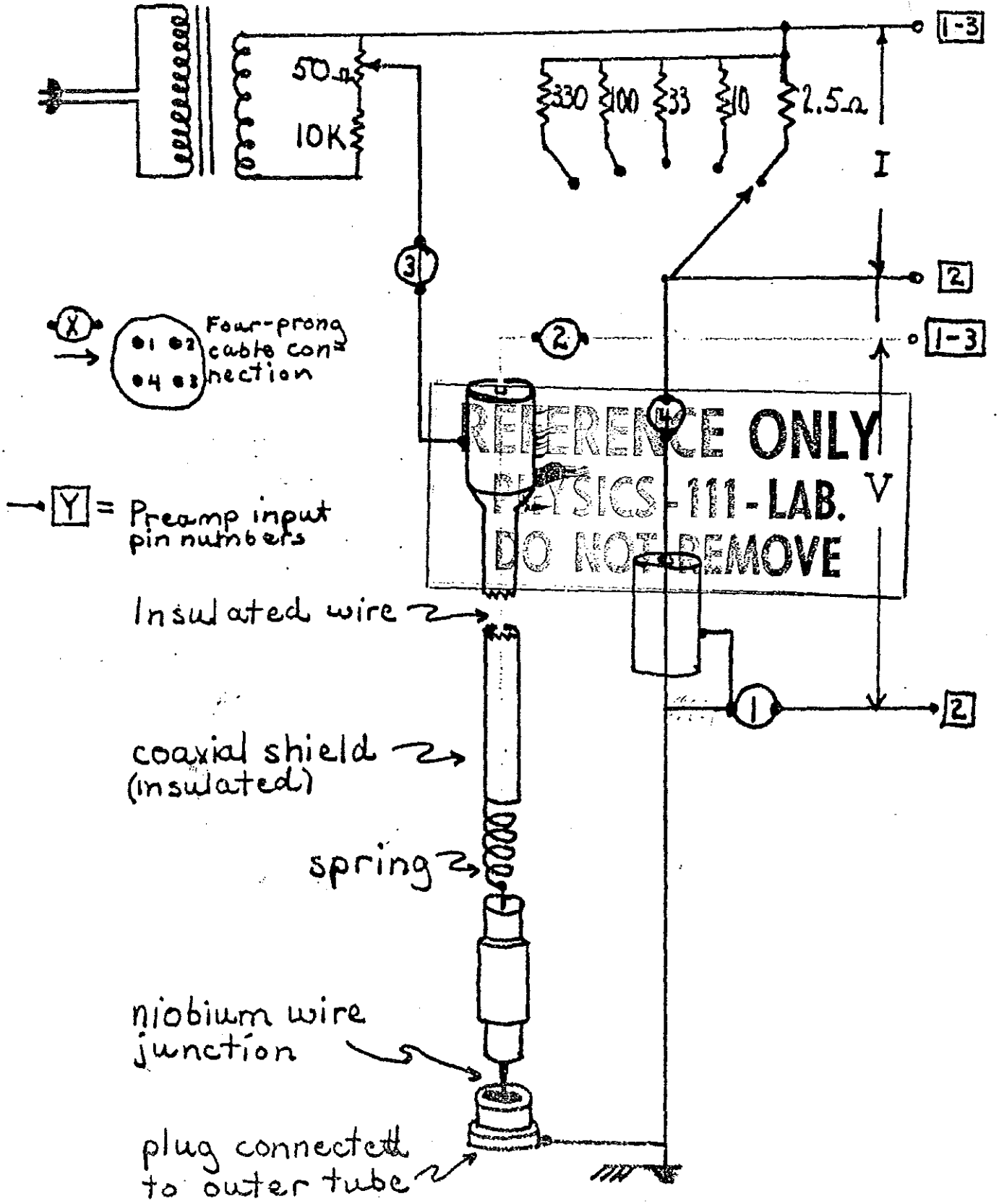
6. The dewar flask has a 24 liter capacity with a liquid nitrogen jacket. It may be found on its special carriage outside Rm 103 LeConte. When the probe is inserted the only opening should be the rubber slit valve as water in the air will freeze the probe tight.

7.-8. The probe is a long coaxial tube with electrical connections for power supply and microwave oscillator. At the bottom is a point-contact niobium wire junction. Microwaves are weakly coupled to the junctions due to leakage down the open pipe.



The power supply generates 60 Hz current for the V-I plot. Electrical connections are shown schematically below.

Electrical Connections for the Probe.



Electrical Connections

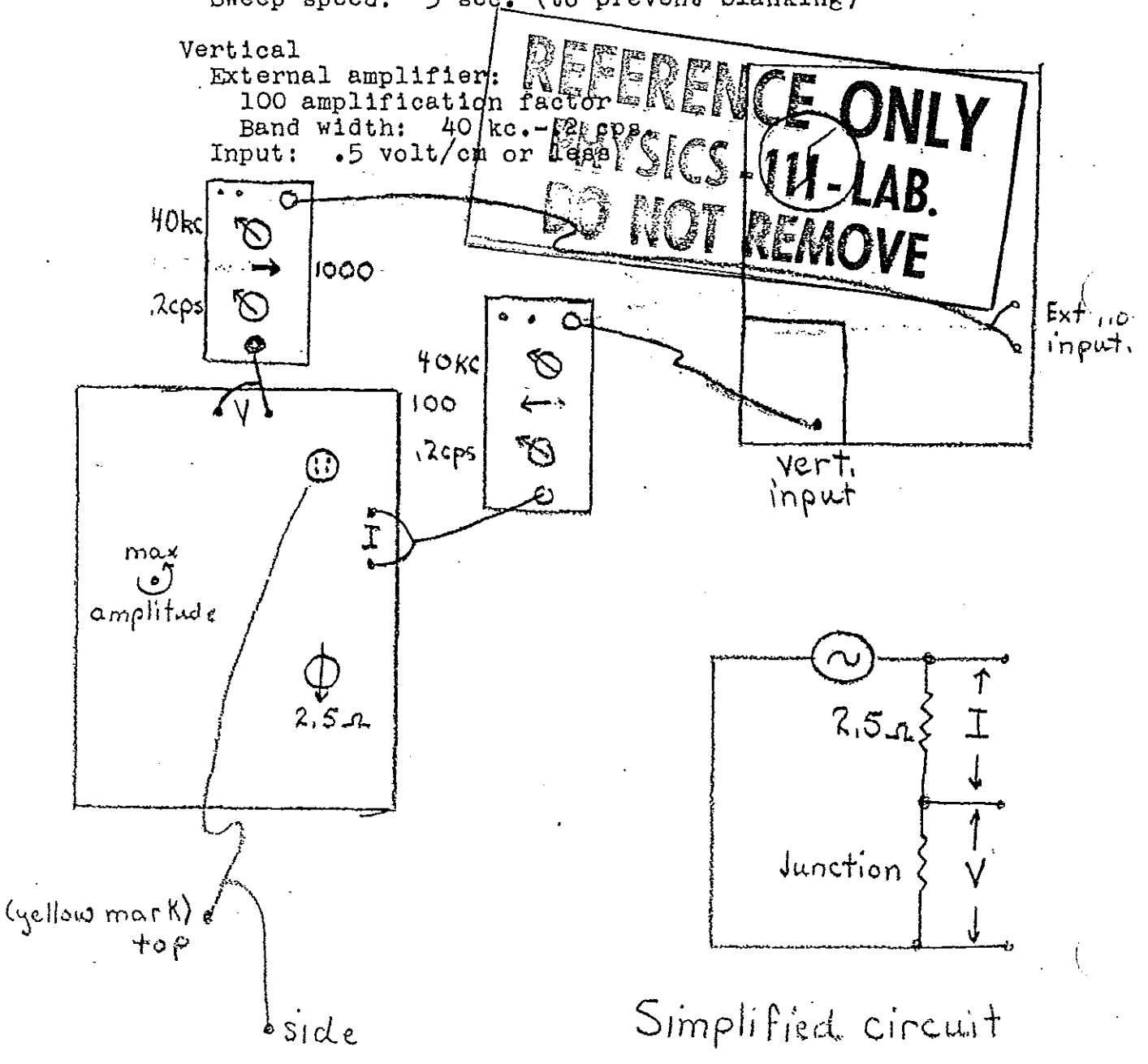
Because of the high cost of liquid helium ($8 per liter) the external electronics must be in order before the probe is cooled.

Initial scope settings

Horizontal-external sweep
External amplifier:
1000 amplification factor
Band width: 40 kc.-.2 cps.
External sweep attenuator: 1 volt/cm.
Sweep speed: 5 sec. (to prevent blanking)

Vertical
External amplifier:
100 amplification factor
Band width: 40 kc.-.2 cps.
Input: .5 volt/cm or less

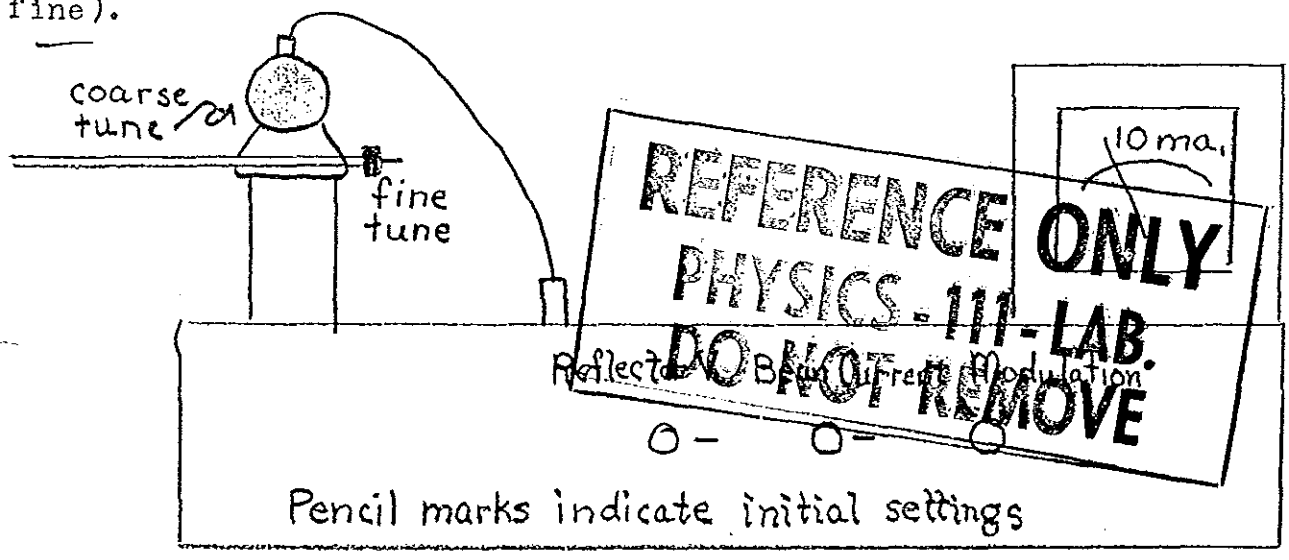
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Simplified circuit

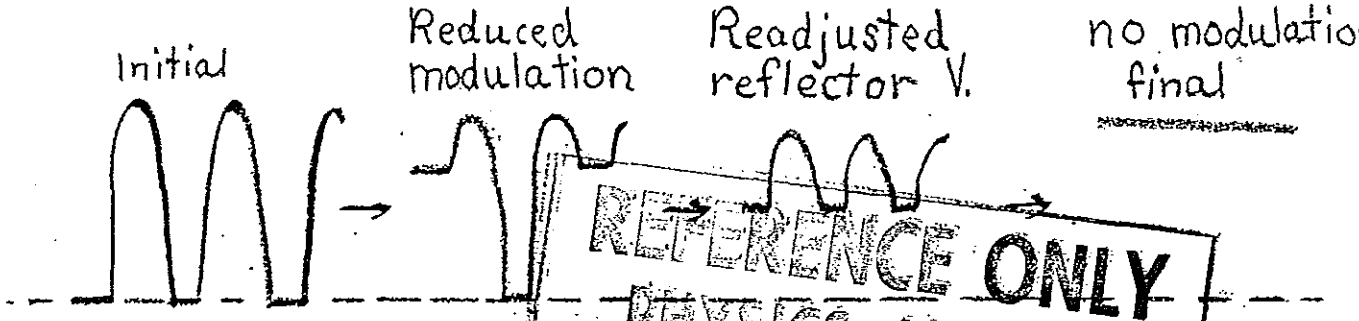
Adjustment of Microwave Generator

There are five parameters involved in the adjustment of the microwave klystron oscillator: reflector voltage, reflector voltage modulation, beam current, and cavity volume (coarse and fine).



The knob marked "modulation" imposes a sixty-cycle signal on the reflector voltage, thus frequency-modulating the microwave carrier. The detector is a crystal half-wave rectifier which should be connected to the vertical input of the scope (sweep speed--2 milliseconds). The beam current is operated normally at 10 ma. (12 ma. max.) and the cavity tuning knob 3 full turns in from where the screw first touches the lever. When reflector voltage is correct one will obtain a set of evenly spaced humps symmetric about the maximum. As the modulation is reduced to zero, the flat portions of the trace will rise symmetrically about the maxima; if not, then the reflector voltage is incorrect. It is best to adjust both controls simultaneously. When properly adjusted for maximum power, the detector produces one volt

~~We have found the second reflector voltage peak (and the third or fourth cavity size peaks) to be strongest~~. In the experiment the modulation is left off.

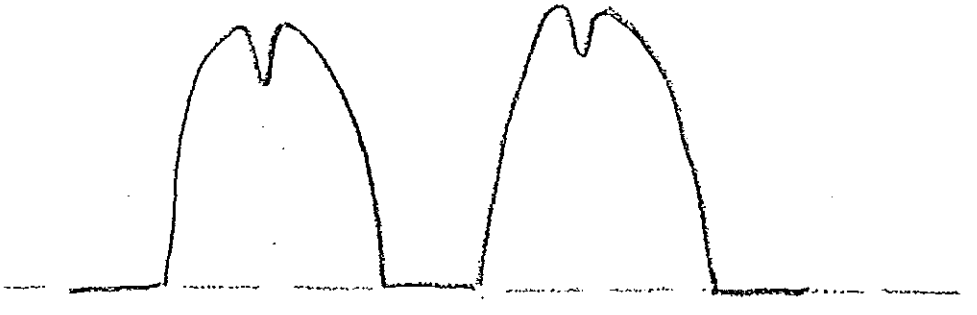


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Use of Wave-meter

The wave-meter is a cavity placed in front of the reflector. When tuned to the proper wavelength it will absorb power and thus produce a pip on the trace as the modulated reflector voltage sweeps past the cavity frequency. When the pip is in the center of the hump (corresponding to the maximum frequency power) the wave meter cavity is in tune. Since in normal operation the modulation is not employed, it should be turned off once the pip is found and the cavity readjusted (the whole line will move.

The present klystron is a K-band oscillator, approximately 24 gigacycles, but the strongest peak is around 22.2 gigacycles. For our data we borrowed a direct-reading model, but it is too expensive for this lab. Another one must be obtained before this experiment may be conducted.



The simplified circuit shows that the voltage across the junction is measured directly by the horizontal amplifier while the current is taken as the voltage induced across a 2.5 resistor in series and displayed on the vertical scale. Where the D.C. resistance of the junction 2.5 and the amplifier equal the trace would appear as a 45 degree line . if contact is not made the trace will be a horizontal line (open-circuit voltage).

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Cooling the Probe

The liquid He dewar may be found on a dolly outside Room 103 LeConte (old section). The rollers should be removed before it is positioned in line with the wave-guide. Before use the entrance must be freed of ice with the hollow aluminum tube that is left on the table.

The probe has an O-ring seal to prevent air from entering the dewar shaft, freezing, and locking the probe in place. If this O-ring seal is held next to the dewar opening, the probe may be <u>slowly</u> pushed into the helium with little waste. If done properly the helium vapor emerges only from the one-way slit in the surgical rubber tubing. It is best to have a TA on hand.

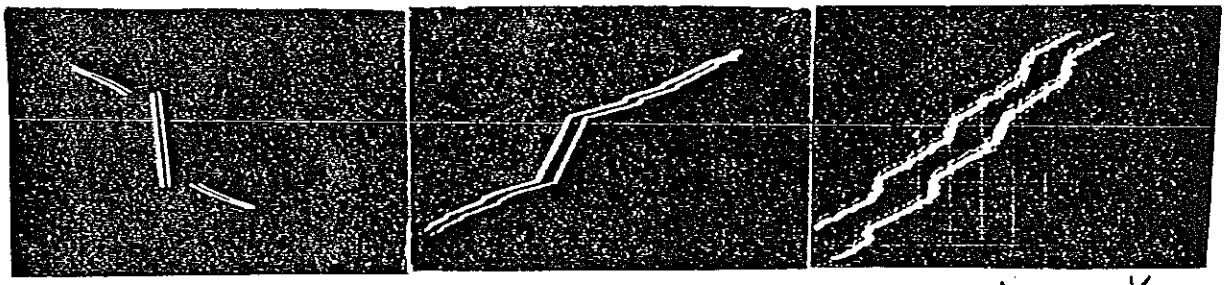
Before the dewar is returned, the volume must be measured. The thin metal tube attached to the dewar has a rubber membrane which vibrates at different frequencies above or below the level of the helium (above--high amplitude 100 cps., below--low amplitude 30 cps.). The graph in the log book converts the measured

distance from ice pack to helium level into liters present in the dewar. With care, waste may be kept below one-half liter; more than 2 liters indicates poor technique.

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Experimental Procedure

When a good Josephson junction has been found (and microwave power applied) the V-I plot reveals a uniform pattern of steps along the circuit load line, the difference between each step being $V = \frac{h\nu}{2e}$.



(reversed) #1 #2 $V = h\nu/2e$ #3

A good junction may be difficult to obtain and it is advisable to have a number of points available. To find the junction the center rod is slowly lowered until there is a jump in the scope trace. The pressure is kept at minimum because the oxide layer may be easily punctured. Once contact is made, the rod is gently twisted. When trace number one (above) is found, microwave power is applied ("100%" on the attenuator means full transmission). Trace number two should appear. If not, there is probably a superconducting short (i.e. the central portion will flatten out due to the critical field applied by the microwaves). The small impedance matching plunger may be pushed in to better couple the microwaves to the junction. Best results occur when the plunger almost

shorts against the central rod (if it shorts the trace becomes a vertical line). It may take some searching to find a suitable junction. Once a trace similar to number two appears, adjustment of the attenuator for maximum transmission and of the plunger for best coupling should produce the desired steps.

For measurement of the voltage between steps the scope is adjusted to include the maximum number of steps in the 10 cm. width and then calibrated, using the calibrated output. A cable has been constructed to go from the calibrated output to the input of the external horizontal amplifier for this purpose.

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DATA, ERROR ANALYSIS AND FURTHER SUGGESTIONS

By

E. I. Caustin, Jr.

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ERROR ANALYSIS

DATA

N	# OF STEPS	CENTIMETERS	$\frac{CM}{STEP} = X$
1	28.5	7	.2456
2	27.5	7	.2545
3	33	8	~~.2424~~
4	31	7	.2484
5	32	7.6	.2375
6	33	8	.2424
7	23	6	.2609
8	28	7	.2500
9	29	7	.2414
10	29	7	.2414
11	20	5	.2500
12	32.5	8	.2462
13	11.5	3	.2609
14	14.	4	.2857 *

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$\Sigma = 3.2216$

$n = 13$ *

$$\bar{X} = \frac{3.2216}{13} = .2478$$

* THE FOURTEENTH DATA POINT IS DISCARDED BECAUSE IT IS APPROXIMATELY FIVE STANDARD DEVIATIONS [illegible] THE AVERAGE

COMPUTATION OF STANDARD DEVIATION

N	$(\delta = x - \bar{x}) \times 10^4$	$\delta^2 \times 10^8$
1	- 22	484
2	+ 67	4489
3	- 54	2916
4	+ 6	36
5	-103	10609
6	- 54	2916
7	+131	17161
8	+ 22	484
9	- 64	4096
10	- 64	4096
11	+ 22	484
12	- 16	256
13	+131	17161

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$$\sum \delta^2 = 65188 \times 10^{-8}$$

$$\sigma = \sqrt{\frac{\sum \delta^2}{n-1}} = (6.52)^{\frac{1}{2}} (.289) \times 10^{-2}$$

$$= 7.4 \times 10^{-3}$$

$$X = \frac{cm}{STEP} = .2478 \pm .0074 \frac{cm}{STEP}$$

CALIBRATION OF OSCILLOSCOPE SCALE

$$\frac{2\ mV}{9.05\ cm} = .2210\ \frac{mV}{cm} = 2.210 \times 10^{-4}\ \frac{volt}{cm}$$

COMPUTATION OF VOLTS / STEP = ΔV

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$$\Delta V = .2478\ \frac{cm}{STEP} \times 2.210 \times 10^{-4}\ \frac{VOLTS}{cm} = 5.476 \times 10^{-5}\ \frac{VOLTS}{STEP}$$

FREQUENCY MEASUREMENT

$\nu = 22.28 \times 10^{9}$ cps (MEASURED THREE TIMES)

How?

COMPUTATION OF $\frac{\nu}{\Delta V}$

$$\frac{22.28 \times 10^{9}}{5.476 \times 10^{-5}} = 4.069 \times 10^{14}\ \frac{CPS}{VOLT}$$

How?

COMPUTATION OF THEORETICAL VALUE

$$\frac{2e}{h \cdot 300}\ ^{*} = \frac{2 \times 4.803 \times 10^{-10}\ esu}{6.625 \times 10^{-27}\ erg\text{-}sec \quad 3 \times 10^{2}}$$

$$= 4.833 \times 10^{14}\ \frac{CPS}{VOLT}$$

* THE 300 IS A FACTOR USED TO CHANGE FROM GAUSSIAN TO PRACTICAL UNITS

COMPARISON OF THEORETICAL AND EXPERIMENTAL VALUES

$$\begin{array}{r} 4.833 \\ 4.069 \\ \hline .764 \end{array} \qquad \frac{.764}{4.833} \times 100 = 15.8\% \text{ DISCREPANCY}$$

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For experimental values measured directly see W.H. Parker, B.N. Taylor & D.N. Langenberg "Measurement of 2e/h Using the ac Josephson Effect ..." Physical Review Letters 18, 287 (Feb. 20, 1967)

Their value is

$$\frac{2e}{h} = 483.5912 \pm .0030 \quad \frac{\text{Mega Hertz}}{\text{micro-volt}}$$

(one Mega Hertz = one kilo mega cycle = 10^9 cps)

ERROR ANALYSIS

see comments on last page for what I consider source of error. -gs

The discrepancy between the experimental and the theoretical value of $\frac{e}{h}$ is too high to be attributed to the measuring instruments or their calibration. These latter errors would be approximately: 1% at most for the wavemeter, 1-5% error for the calibrated voltage output on the oscilloscope and the measurement of that instument's horizontal gain. These together would ~~seem to~~ contribut approximately 2-5% error, not nearly the discrepancy we are faced with. It should be pointed out that the completion of several more runs of the experiment would reduce these errors by means of their statistical nature. However, we were unable to complete more than a single run of the quantitative experiment.

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The day after this single run was made, one of the experimenters found that a wire in the probe structure had been broken, and that the top and bottom segments of this wire were uninsulated and were ~~in~~ twisted. It could be inferred that this wire was broken during the experimental measurement by the fact that it was very difficult to achieve the s-c Josephson pattern on the screen. At the time of the measurements, however, this difficulty was attributed to the difficulty of finding the proper spot for the probe tip on the flattened niobium wire. Undoubtedly, more experience with the junction would have enabled us to more quickly

suspect the broken wire as the trouble.

The pertinence of this broken wire is that if it were to have come in contact with the stainless steel tube when the experiment was being carried out, there would have been a small resistance of the current circuit placed in series with the voltage leads. (See diagram #1.) This would have the effect of causing a slight slope in the zero resistance vertical portions of the Josephson step curve. A difference in the slope from vertical of 10-15% would not have been immediately visible due to the small absolute height of the steps. Unfortunately for the two reasons the slope was not measured on a higher scale of oscilloscope amplification. This omission was due to two reasons. Firstly, when the Josephson pattern with its apparently vertical, zero resistance step, had been produced on the oscilloscope screen, it was wrongly assumed that the probe was operating properly. Secondly, and most frustrating was, that we ran out of time to complete and analyze all of the data we were hoping to. It may be suggested that future workers on this experiment, after obtaining the Josephson step curve, should immediately check the slope of the vertical step segment on a high gain of the scope in order to spot the possibility of a shorted center wire which would be indicated by a non-vertical step. What makes this interpretation of the discrepancy even more reasonable is the fact that when Dr. Richards found that the wire had broken once before, he "made-shifted" a jumper which connected the voltage

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lead to the current source output electrically before the latter was even connected to the probe. This had the effect of introducing a severe change in the slope of the zero resistance step segment.

SUGGESTIONS

1. The center wire can be replaced with some sort of flexible and better insulated (present wire uses brittle lacquer insulation) center wire. Dr. Richards suggested one type of flexible wire; however, it is uninsulated and thus might present a problem. There should be some kind of wire to fill these conditions, however it might be necessary to purchase it specially.

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2. Supporting blocks for the waveguide which goes from the klystron oscillator to the probe should be built in order to avoid the necessity of "male-shifting" a support with the inherent dangers of letting the waveguide slip and thus damaging the klystron.

3. A good wavemeter of the type found in Jeffries' and Kittel's group should be gotten especially for use in this experiment. The home-built waveguide we were originally furnished with did not cover the range in which we had to operate. Our measurements were taken at 22.28 Gc and the range of the klystron is about 21-25 Gc. Since frequency measurement accuracy is of high importance, it should be a reliable wavemeter.

4. It should be pointed out that on the older of the two Tektronix Preamplifiers (the one without the on-off panel light) the D-C level control does not function, and the amplification is approximately 4000. According to Dr. Richards these amplifiers have been modified in order to cut down the signal-to-noise ratio. One of the modifications could have eliminated the D-C level control, but the cause of four- fold again is unknown.

5. It is necessary to recognize at the start that both preamplifiers must be operated with the highest possible bandwidth (highest high-frequency cut-off and lowest low frequency cut-off). This is due to the high bandwidth of Fourier components necessary to construct the vertical to horizontal step structure of the Josephson curve.

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6. The oscilloscope needs to be repaired. a) The left channel does not seem to operate at all. b) The triggering controls do not appear to function on any of the triggering selection positions. c) The horizontal sweep is intermittant.

7. A brief suggestion for an interesting possible experiment with the double Josephson junction can be made. A plug for the bottom of the probe could be constructed with two flattened pieces of niobium wire, and a moveable plug could be fitted with two niobium wire points and a set of wires to create a magnetic field. (See diagram #2.) In this way there would be a finite measurable area between two Josephson junctions, and with use of the variable magnetic fields from the fixed wires, the effect of flux

...ization should be more easily measureable. The most difficult part would be the geometry of the magnetic field wires (made difficult by the constriction of getting the probe through the narrow neck of the liquid helium dewer neck) and the computation of the magnetic flux through the effective area of the double-junction. However, these problems with some faculty help would not be unsurmountable difficulties.

E. There was a double trace on the oscilloscope of the Josephson curve. This can be accounted for by the phase shift caused by the leads' and probe's stray inductances and capacitances. Although this double trace does not hinder measurement appreciably, esthetically it would be nicer to have a single trace. This could be done by placing the appropriate inductance or capacitance in the circuit to create another intentional phase shift in the opposite direction of the inherent one. Probably the experimental use of a capacitance and inductance substitution box or their equivalents would quickly lead to the appropriate values and placement of this phase shifter.

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Not true. This ac pickup is a serious source of systematic error. It adds to the true voltage across the junction an additional voltage which is not constant, not even linear, with time, and with arbitrary phase shifts. This means that the spacing between Josephson jumps is different for upper and lower traces of the loop -- different even between different jumps on the same trace.

1) Efforts should be made to eliminate this pickup.

2) Averages of the jump spacing must be made over the

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THE JOSEPHSON EFFECT

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Physics 111
June 6, 1967

JOSEPHSON EFFECT THEORY

By

Ron Anderson

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Introduction

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The most exciting aspect of the Josephson effect experiment is that it allows an undergraduate to duplicate work which was only theory in 1962. Compared to most of the experiments available in the 111 lab, this is much more exciting because it is so modern, and because it allows the student to measure the value of e/h in a macroscopic manner. In order to understand the experiment, it is necessary to have a working knowledge of quantum mechanics. The Josephson effect consists of quantum mechanic electron pair tunneling. These electron pairs behave as bosons, whereas normal single electron tunneling is described by fermi particle characteristics.

It's not all that ominous.

A Josephson junction usually consists of two pieces of superconductor between which is sandwiched a very thin dielectric. The dielectric is conveniently chosen to be the oxide of the superconducting metal. Earlier forms of the junction were made by vacuum deposition of a metal as a strip on a glass slide, allowing it to oxidize, and then vacuum depositing another strip of metal at right angles to the first. The area where the two strips crossed formed the junction. While this method is still used to produce the best junctions, the technique is exceedingly difficult and tedious. The point junction used in this experiment seems much easier to produce and also allows better coupling for the microwaves than the thin film junction. Bridge junctions consisting of two areas of metal film connected by a very narrow strip of film(a few microns wide) have also been used to produce the Josephson

effect.

The Josephson effect can be conveniently divided into three parts: if no potential bias is placed on the junction, a finite current is observed to flow in the junction - this is called the D. C. Josephson effect; if the junction is now biased with a direct current, it produces radio frequency current oscillations called the A. C. Josephson effect; the third is the quantum interference effect in which magnetic flux passing through the area of two or more Josephson junctions in parallel causes the supercurrent to undergo oscillations with quantized changes in the flux.

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Superconduction by Electron Pairs

Metals exhibit resistance to the flow of current due to scattering from vibrating atoms and electrons or from crystaline imperfections and impurities. Superconductors, however, permit free electrons to move in unison through the metal crystal without resistance due to scattering. An electron moving through a superconductor distorts the lattice by attracting positive metal ions in its wake. The concentration of positive charges attracts other electrons so that the net effect is an attraction between electrons. Metals which conduct well at normal temperatures do so because their free electrons have a very weak interaction with the crystal lattice. The weak interaction reduces the scattering of single electrons in normal conductors, but it also reduces the net attraction between electrons which superconductors possess.

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As a result of the interaction, two electrons of equal and opposite spin and momentum are coupled together in weakly bound pairs. They are so weakly bound, that at a few degrees above absolute zero Kelvin, the thermal agitation breaks the bond and destroys superconductivity. Compared to the distance between the lattice ions, the two paired electrons are separated thousands of times farther apart because they are so weakly bound. The Pauli spin principle of quantum mechanics states that two electrons of the same spin can not occupy the same position, and the electron pairs around each ion usually overlap with millions of other pairs. This causes the motions of all the pairs to be correlated. As a result, the center of mass of each pair moves with the same momentum as if all the pairs were locked in unison. They share the same quantum mechanical phase relationship and satisfy the characteristics of bosons.

Phase Relationship in Superconductors

If a voltage V is applied across the ends of a superconducting wire, an electron pair at one end of the wire will have an energy greater by 2eV than a pair at the opposite end. The difference in energy states would produce a phase change throughout the wire and breakdown the electron pair phase bond. This would cause the whole wire to be in a higher energy state. However, if the potential difference across the wire does not exceed a critical point, the wire will keep its lower energy state by conducting current, thereby eliminating the potential across it.

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In the Josephson junction, the phase of electron pairs will be the same in each of the two pieces of superconductor, but the phases of the two superconductors may not be the same. An electron pair which tunnels through the insulator will change phase. If the pair has the same phase upon reaching the other superconductor, it will be successfully accepted there and contribute to a current. The net current will be zero if an electron pair traveling in the opposite direction also matches phase. However, an electron pair which does not match phase after tunneling will be reflected back to the superconductor from whence it came. Thus there will be a supercurrent through the junction when electron pairs succeed in matching phase in one direction, but not in the other. The magnitude of the current has a value of $J_o \sin \delta$, according to Josephson's theory, where δ is the phase difference.

The phase difference can be controlled in the laboratory by adjusting the current that flows through the junction. The phase difference is also directly affected by an external magnetic vector potential. The change of phase in the insulating barrier varies as the magnitude of a magnetic field applied at a right angle in the plane of the barrier.

Since the current is a function of the phase difference and the phase difference is a periodic function of the magnitude of the magnetic field, the current also varies periodically as the magnitude of the magnetic field is changed. Using this fact, John Rowell verified the existence of the D. C. Josephson effect in 1963. He observed that when the magnetic field was slowly increased from zero, the Josephson current went to zero at a value of magnetic field which can be labeled H_o. With a further increase of the flux, the current reached a maxima at $3H_o/2$, $5H_o/2$, etc., and went to zero at $2H_o$, $3H_o$, etc.

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If a current source is attached to the junction and the current is increased above the value of the Josephson current for that particular junction, a potential will be developed across the junction. This potential causes a change in the phase difference across the junction which increases steadily in time. The phase conditions for electron pair transfer through the insulator are constantly changing. First the electron pairs are able to transfer only one direction; the next moment, they are able to transfer only in the opposite direction. This means that supercurrent oscillates back and forth at a very high rate within the junction. The frequency of oscillation is directly proportional to the voltage across the junction. In fact, the frequency is given by $\gamma = \frac{2eV}{h}$.

Simple Quantum Mechanical Derivation

The original paper by Josephson in which he analyzed the possibilities of quantum mechanical electron pair tunneling in superconductors is extremely hard for a student to understand. A much clearer explaination is presented in the last chapter of Feynman's Vol. III. The following explaination is derived from this source and from Kittel's notes on Solid State Physics.

D. C. Josephson Effect

Consider a simple Josephson junction where there is a wave function ψ_1 associated with one side of the junction and a function ψ_2 with the other. If the superconductors are identical and at the same potential, the quantum equations for the motion of electrons in the conductors are:

$$i\hbar\dot{\psi}_1 = \hbar T \psi_2 \quad ; \quad i\hbar\dot{\psi}_2 = \hbar T \psi_1$$

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in which the transfer interactions of the electron pairs is represented by $\hbar T$. As the insulating layer becomes thicker than about 10 Å, T goes to zero and ther is no electron pair? tunneling.

Now if we substitute for ψ_1 and ψ_2 the values:

$$\psi_1 = \sqrt{\rho_1}\, e^{i\theta_1} \quad ; \quad \psi_2 = \sqrt{\rho_2}\, e^{i\theta_2}$$

where ρ is the electron density and θ is the phase, then

$$\dot{\psi}_1 = \tfrac{1}{2}\rho_1^{-1/2} e^{i\theta_1}\dot{\rho}_1 + i\psi_1\dot{\theta}_1$$

and

$$i\hbar\dot{\psi}_1 = \tfrac{1}{2} i\hbar\rho_1^{-1/2} e^{i\theta_1}\dot{\rho}_1 - \hbar\psi_1\dot{\theta}_1 = \hbar T \rho_1^{1/2} e^{i\theta_2}$$

Multiply through by $\hbar^{-1}\rho_1^{1/2} e^{i\theta_1}$ to obtain:

$$\tfrac{1}{2} i\dot{\rho}_1 - \rho_1\dot{\theta} = T(\rho_1\rho_2)^{1/2} e^{i\delta}$$

where $\delta = \theta_2 - \theta_1$

Now if we equate the real and the imaginary parts of the above equation:

$$-\rho_1 \dot{\theta}_1 = T(\rho_1\rho_2)^{1/2}\cos\delta \quad ; \quad \dot{\theta}_1 = -T\left(\frac{\rho_2}{\rho_1}\right)^{1/2}\cos\delta$$

$$\tfrac{1}{2} i\dot{\rho} = T(\rho_1\rho_2)^{1/2}\sin\delta \quad ; \quad \dot{\rho}_1 = 2T(\rho_1\rho_2)^{1/2}\sin\delta$$

Under the conditions we have stated, $\rho_1 = \rho_2 = \rho_0$, so: $\dot{\theta}_1 = \dot{\theta}_2$

and $$\dot{\rho}_2 = -2T(\rho_1\rho_2)^{1/2}\sin\delta$$

Now, the current through the junction must be proportional to $\dot{\rho}_1$ or $-\dot{\rho}_2$ so:

$$J = J_0 \sin\delta$$

where

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or J_0 is dependent on T, a characteristic of the junction. With zero potential, the current through the junction will have a value between J_0 and $-J_0$ depending on the value of δ, the phase difference.

A. C. Josephson Effect

If the junction is now biased with a D. C. voltage, the equations of motion must be rewritten to give:

$$i\hbar\dot{\psi}_1 = -eV\psi_1 + \hbar T\psi_2 \quad ; \quad i\hbar\dot{\psi}_2 = eV\psi_2 + \hbar T\psi_1$$

And, following the same steps as above:

$$\tfrac{1}{2} i\hbar\rho_1^{-1/2} e^{i\theta_1}\dot{\rho}_1 - \hbar\psi_1\dot{\theta}_1 = -eV\rho_1^{1/2}e^{i\theta_1} + \hbar T\rho_2^{1/2}e^{i\theta_2}$$

or $$\tfrac{1}{2} i\dot{\rho}_1 - \rho_1\dot{\theta}_1 = -eV\rho_1\hbar^{-1} + T(\rho_1\rho_2)^{1/2}e^{i\delta}$$

then: $$\dot{\rho}_1 = 2T(\rho_1\rho_2)^{1/2}\sin\delta \quad ; \quad \dot{\delta} = \left(\frac{eV}{\hbar}\right) - T\left(\frac{\rho_2}{\rho_1}\right)^{1/2}\cos\delta$$

also,

$$\tfrac{1}{2} i\dot{\rho}_2 - \rho_2\dot{\theta}_2 = eV\rho_2\hbar^{-1} + T(\rho_1\rho_2)^{1/2}e^{-i\delta}$$

$$\dot{\rho} = -2T(\rho_1\rho_2)^{1/2}\sin\delta \quad ; \quad \dot{\delta} = -\left(\frac{eV}{\hbar}\right) - T\left(\frac{\rho_1}{\rho_2}\right)^{1/2}\cos\delta$$

Again, $\rho_1 = \rho_2 = \rho_0$ and using this fact in the values of $\dot{\epsilon}_1$ and $\dot{\epsilon}_2$, we find that:

$$\dot{\delta} = -\frac{2eV}{\hbar}$$

so

$$\delta = \delta_0 - \frac{2eVt}{\hbar}$$

and the current is given by:

$$J = J_0 \sin\left[\delta_0 - \frac{2eVt}{\hbar}\right]$$

oscillating with a frequency ω: $\omega = \frac{2eV}{\hbar}$

This means that everytime an electron pair tunnels through the insulating barrier, a photon of energy $\hbar\omega = 2eV$ must be emitted or absorbed.

By applying both a direct current bias and a small a. c. voltage with a very high frequency, we obtain:

$$V(t) = V_0 + v\cos\omega t$$

which makes $\delta(t)$ thus:

$$\delta(t) = \delta_0 - \frac{2eVt}{\hbar} - \frac{2ev}{\hbar\omega}\sin\omega t$$

then

$$J = J_0\left\{\sin\left[\delta_0 - \frac{2eVt}{\hbar}\right] - \frac{2ev}{\hbar\omega}\sin\omega t\cos\left[\delta_0 - \frac{2eV_0t}{\hbar}\right]\right.$$

The first term on the right has a time average value of zero, but the second term does not if $\omega = \frac{2eV_0}{\hbar}$ so, there will be a direct current through the junction if a small $(v/V_0 \ll 1)$ rf voltage is induced across it with frequency $2eV_0/\hbar$.

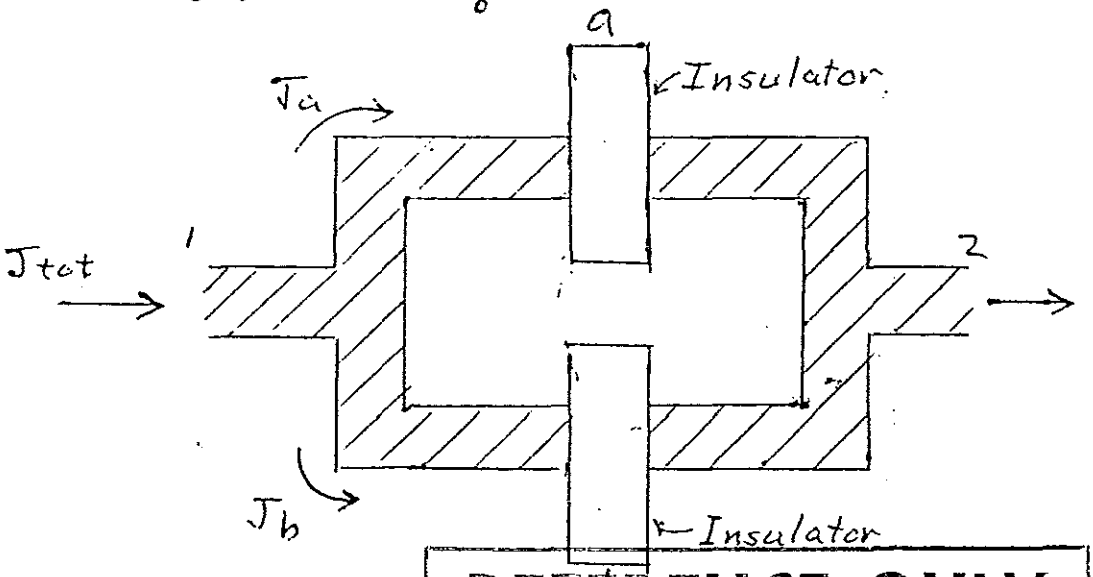
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Macroscopic Quantum Interference

The phase difference δ around two Josephson junctions connected in parallel which encompass a magnetic flux $\phi = \oint A \cdot d\ell$ is:

$$\delta = \frac{2e}{\hbar c}\oint A \cdot d\ell$$

The following refers to the schematic diagram of a double junction seen below. The phase difference of current through junction a is δ_a and through junction b is δ_b .



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When flux passes through the interior of the above circuit, it changes the phase relationship of the junctions so that :

$$\delta_b - \delta_a = \frac{2e}{\hbar c}\Phi$$

or

$$\delta_b = \delta_0 + \frac{e}{\hbar c}\Phi \quad \text{and} \quad \delta_a = \delta_0 - \frac{e}{\hbar c}\Phi$$

$$J_{tot} = J_a + J_b = J_0\left\{\sin\left(\delta_0 + \frac{e}{\hbar c}\Phi\right) + \sin\left(\delta_0 - \frac{e}{\hbar c}\Phi\right)\right\} = J_0 \sin\delta_0 \cos\frac{e\Phi}{\hbar c}$$

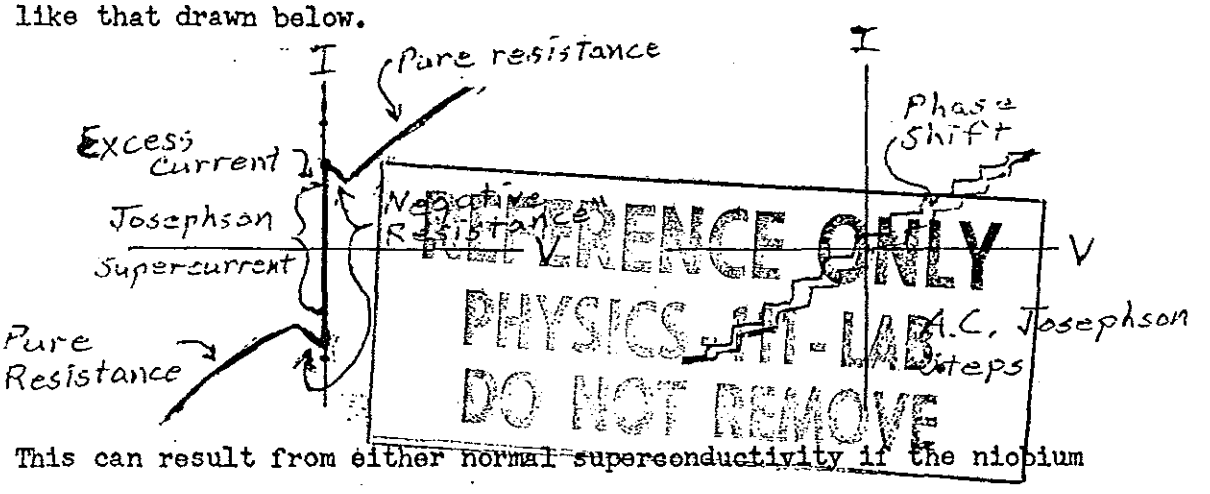
$J(\Phi)$ will have maxima at $\frac{e\Phi}{\hbar c} = n\pi$ where n = integer

or $$\Phi = \frac{n\pi\hbar c}{e} = n\frac{hc}{2e}$$

This effect makes the double Josephson junction useful as a very accurate magnetometer, capable of measuring a field as small as 2×10^{-7} gauss.

Application of Theory to the Experiment

When the point junction is properly adjusted in the absence of external microwaves, it is possible to obtain a current voltage trace like that drawn below.



This can result from either normal superconductivity if the niobium point has pierced the oxide layer or from D. C. Josephson superconductivity. If the trace is due to the D. C. Josephson effect, the vertical slope supercurrent will oscillate in magnitude quite rapidly as a large magnet is moved toward the dewer. The magnet will have no effect on the trace if it has been caused by the niobium wires shorting. In a good point junction, a 2000 gauss magnet will show an observable effect as it is moved at a distance of approximately 10 feet from the dewer. For the D. C. Josephson effect, the magnet causes a phase shift between electron pairs and this changes the current that flows through the junction. Also, due to irregularities in the niobium metal surfaces, there are actually several point junctions at the end of the wire. This means that there will be quantum mechanical interference when magnetic flux passes through the finite areas formed by the sub-points. This was described above for the case of a double point junction, but here we have an effective multiple point junction which produces an equivalent multiple slit interference

pattern in the current as the flux changes.

The pure resistance part of the trace (that part of the trace where there is a definite slope) occurs when the current through the junction exceeds the Josephson supercurrent and a potential appears across the junction. We are using a 60 cycle sine wave to sweep current through the junction, so there are two parts to the resistance curve. It is interesting to note that the vertical current line is often seen to have a value greater than the superconducting current. Then, that part of the trace which goes from the end of the vertical current to the first of the pure resistance indicates a "negative resistance". See figure above.

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The part of the curve which I have called pure resistance is simply where the electron pair phase locking has broken down and single electron tunneling has begun. This trace is also observed in normal superconductivity.

As the microwave power is turned up by adjusting the attenuator, the D. C. Josephson effect breaks up into steps where each of the changes in voltage is equal (see figure). According to the theory, the size of each voltage step is $V = h\nu/2e$ where ν is the microwave frequency. The explaination for this effect is that the voltage which appears across the junction is actually the summation of two voltages. We are sweeping the junction with a 60 cycle voltage and also with a very small voltage which varies at about 22.3 gigacycles per second. The microwave voltage frequency modulates the Josephson supercurrent since the Josephson frequency is a function of applied voltage. The frequency modulation causes the Josephson current to contain components of many frequencies. When the sum of the Josephson frequency plus or minus the microwave frequency plus or minus the component frequencies plus or minus all the

the integral harmonicsof each of these frequencies equals zero, there will be a direct current flow through the junction. Each time all the frequencies add to zero, a step occurs. In simpler terms, the first step occurs when the microwave frequency or an integral multiple of it equals the Josephson frequency. The amplitude of the nth step decreases according to an equation of the nth order Bessel function $J_n(2ev/h\nu_m)$ where v is the voltage of the microwave. If the microwave attenuator is turned completely off (zero attenuation), the steps will sometimes completely disappear. Occassionally, it was possible to observe a step which was twice as long as the others, indicating that single electron tunneling was occurring. Never

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If the point is very carefully adjusted, it will be possible to observe the A. C. Josephson effect without applying external microwaves. The steps are found on the pure resistance part of the trace and result from the Josephson rf produced by the points beating against its own oscillations inside the resonant cavity formed at the end of the probe. These steps have a very small vertical magnitude (current change) because so little power is radiated from the point.

We were bothered by a phase shift in our cables which caused one half of the 60 cycle sweep to be shifted slightly from the other half so that we obtained a double trace. The steps on each of these traces were consistently the same and the phase shift did not effect the experiment, but was rather annoying. Apparently it was caused by impedance and capacitance i the cables.

APPARATUS AND EXPERIMENTAL PROCEDURE

By

W. H. Putnam

Niobium points

1/2" lengths of niobium wire. File with fine-toothed file in drill press at medium speed. Smooth with emery paper. Points must be etched in NaOH (weak soln) to remove iron filings. Leave points for at least a day to oxidize. When a point has been blunted in experiment, it cannot be drawn again as only a limited supply of niobium wire is available. Best shape is like a phonograph needle; narrow but slightly rounded at the end.

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Apparatus

1. Oscilloscope
2. Low-level preamplifiers
3. K-band microwave oscillator
4. Detector
5. Wave-meter
6. Dewar flask
7. Josephson junction probe
8. Power supply for probe

1. The oscilloscope (Tektronix Type 543B with dual-trace plug-in unit--U.C. code 65-10-07405) is used to make an X-Y plot of current vs. voltage. This particular one has been calibrated to 100.00 .01 volts.

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One of these amps is borrowed from Rad lab and must be returned. Order a new one.

2. The preamplifiers (Tektronix Type 122) are wide-band (40 kHz-.2 Hz) differential input amplifiers. In operation one input (pin 1) is grounded to pin 3. The two amplifiers are to provide the necessary .1 mv/cm. sensitivity. Measured amplification factors are 390-3900 horizontal and 142-1420 vertical.

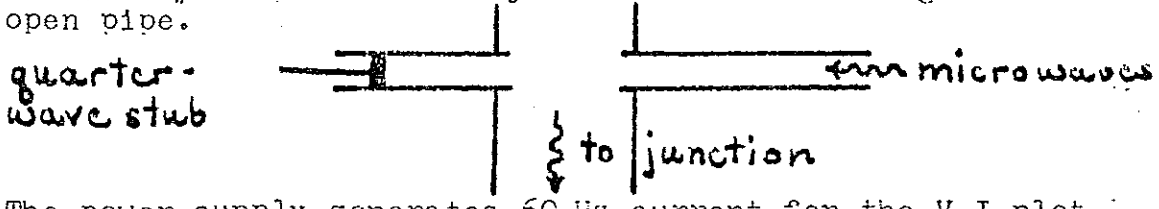
3. Microwave power is supplied by a K-band klystron oscillator (nominal output 20 mw at 22.2 GHz.) The power is fed into a directional coupler (with detector), attenuator, and quarter-wave stub to provide weak coupling of microwaves to the junction at the bottom of the coaxial probe.

4. The crystal detector is an F-R Type K205BF with 1 volt output at maximum klystron power.

5. We have no wave meter at the present time (the former one was outside the range of the klystron).

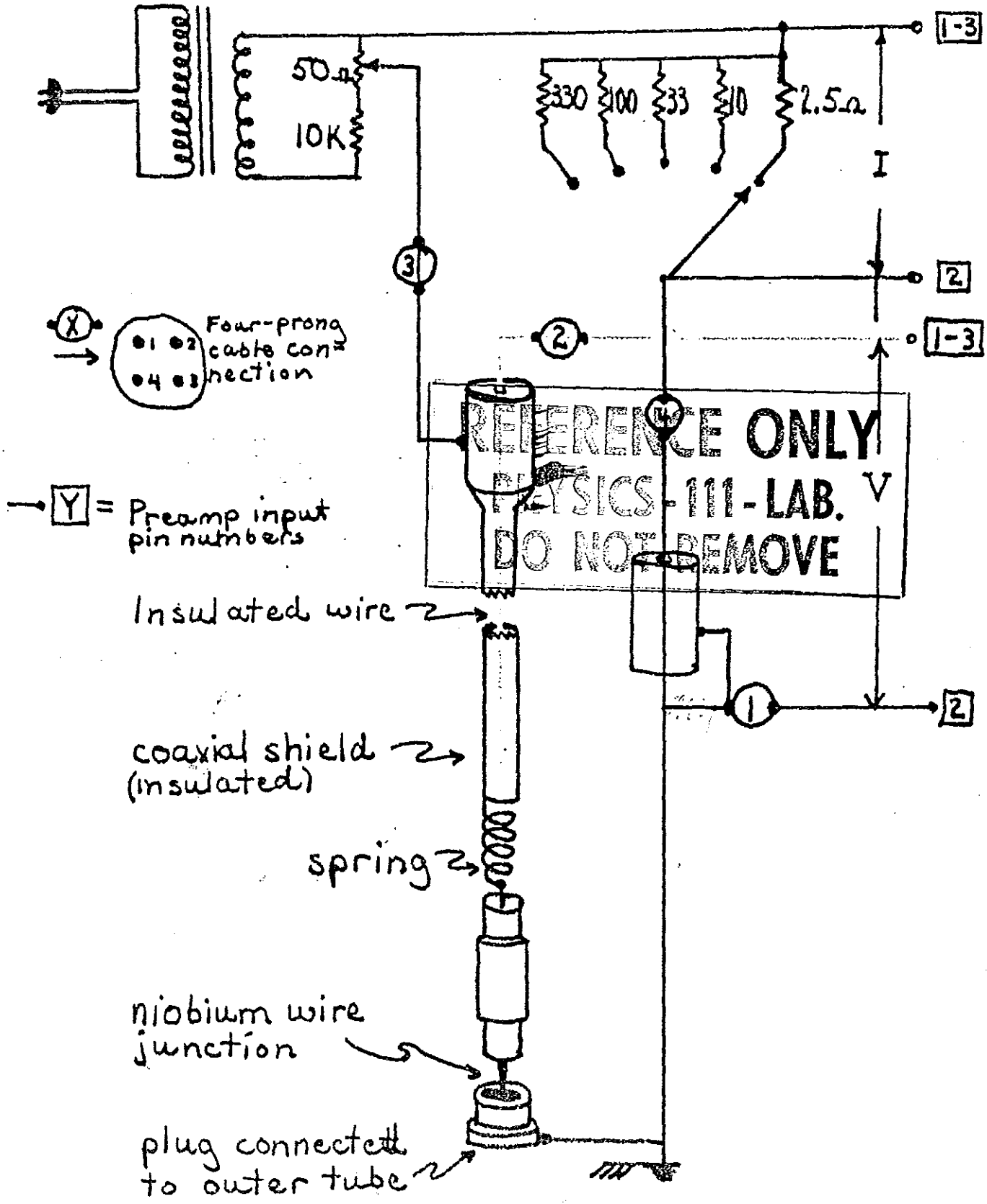
6. The dewar flask has a 24 liter capacity with a liquid nitrogen jacket. It may be found on its special carriage outside Rm 103 LeConte. When the probe is inserted the only opening should be the rubber slit valve as water in the air will freeze the probe tight.

7.-8. The probe is a long coaxial tube with electrical connections for power supply and microwave oscillator. At the bottom is a point-contact niobium wire junction. Microwaves are weakly coupled to the junctions due to leakage down the open pipe.



The power supply generates 60 Hz current for the V-I plot. Electrical connections are shown schematically below.

Electrical Connections for the Probe.



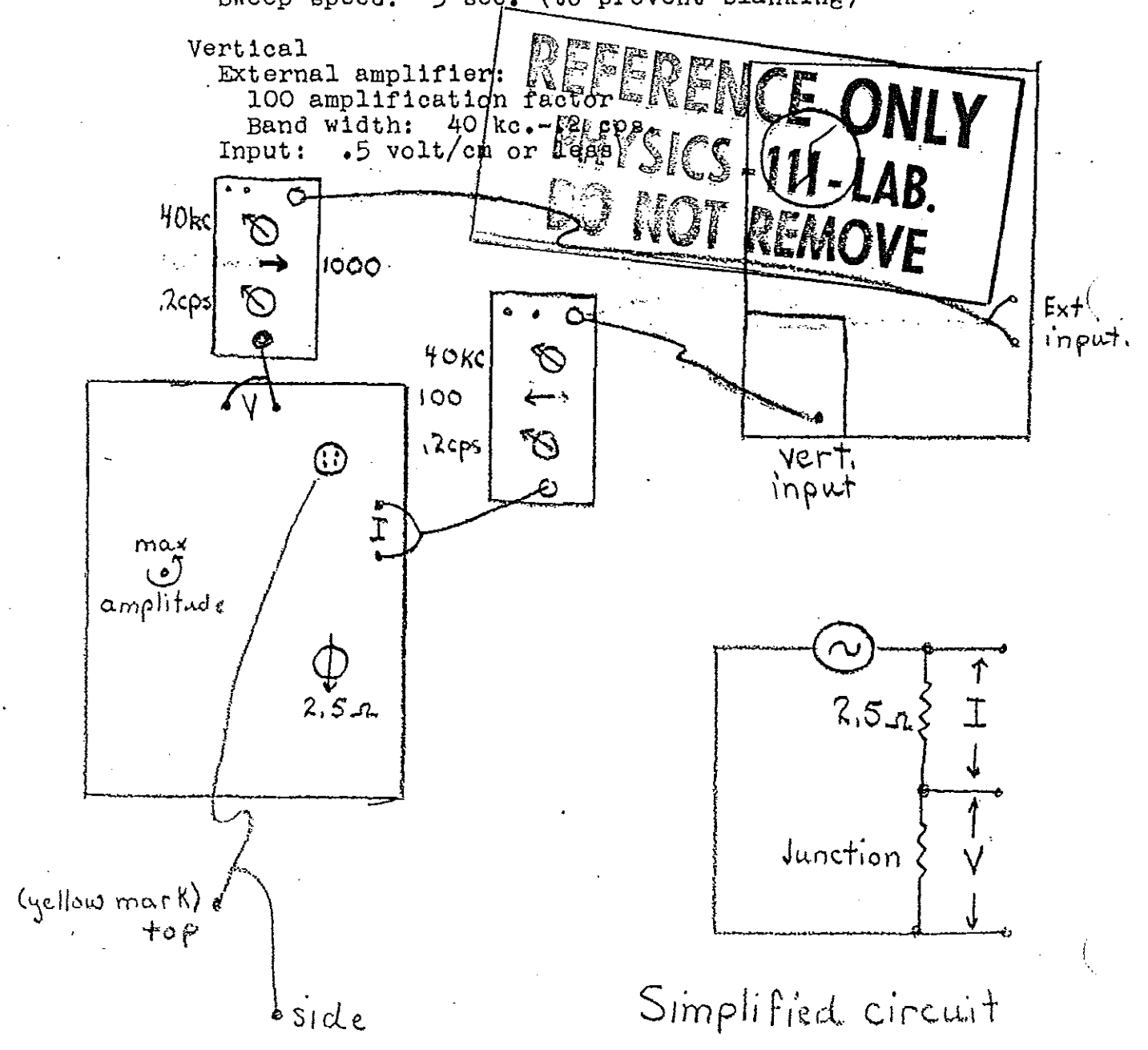
Electrical Connections

Because of the high cost of liquid helium ($8 per liter) the external electronics must be in order before the probe is cooled.

Initial scope settings

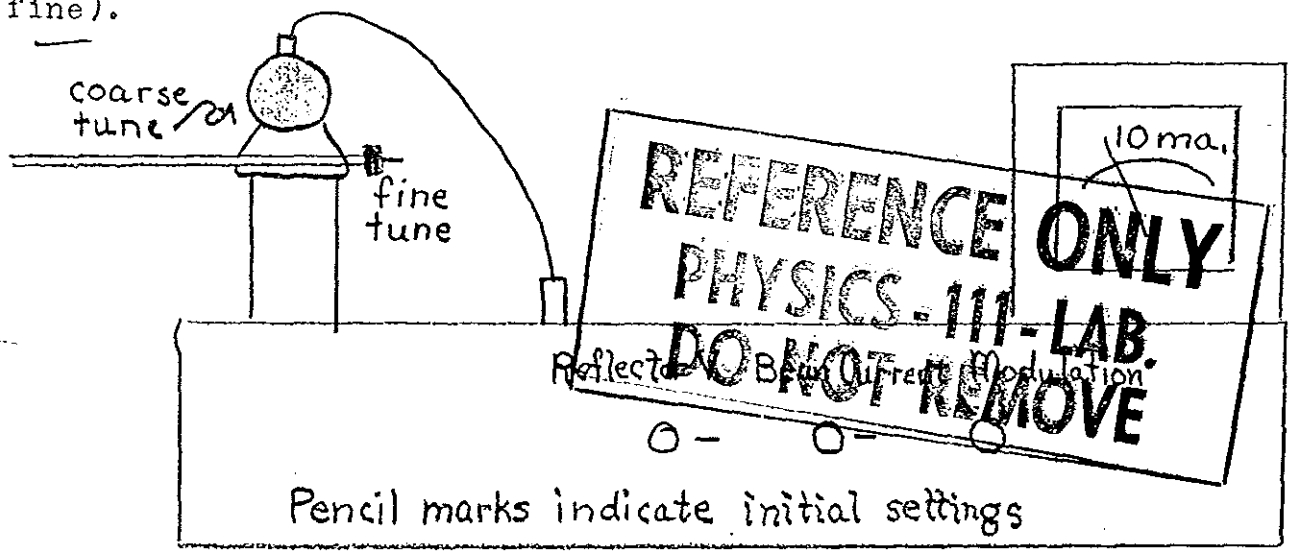
Horizontal-external sweep
External amplifier:
1000 amplification factor
Band width: 40 kc.-.2 cps.
External sweep attenuator: 1 volt/cm.
Sweep speed: 5 sec. (to prevent blanking)

Vertical
External amplifier:
100 amplification factor
Band width: 40 kc.-.2 cps.
Input: .5 volt/cm or less



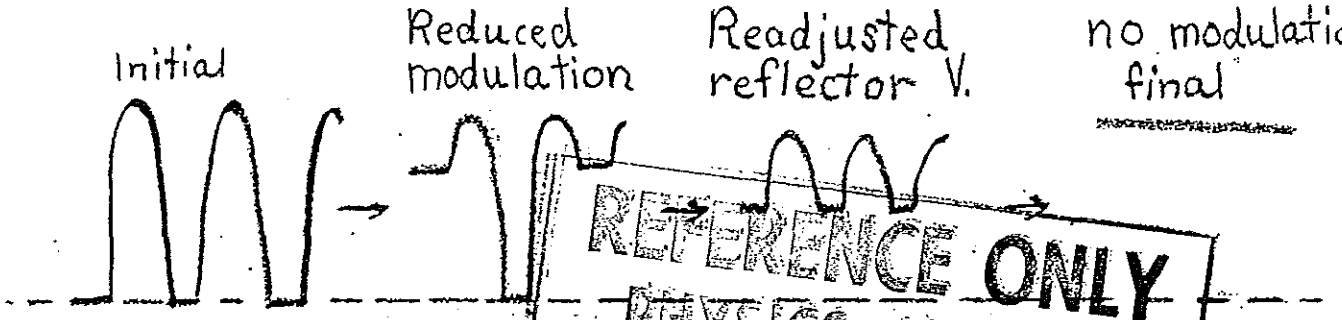
Adjustment of Microwave Generator

There are five parameters involved in the adjustment of the microwave klystron oscillator: reflector voltage, reflector voltage modulation, beam current, and cavity volume (coarse and fine).



The knob marked "modulation" imposes a sixty-cycle signal on the reflector voltage, thus frequency-modulating the microwave carrier. The detector is a crystal half-wave rectifier which should be connected to the vertical input of the scope (sweep speed--2 milliseconds). The beam current is operated normally at 10 ma. (12 ma. max.) and the cavity tuning knob 3 full turns in from where the screw first touches the lever. When reflector voltage is correct one will obtain a set of evenly spaced humps symmetric about the maximum. As the modulation is reduced to zero, the flat portions of the trace will rise symmetrically about the maxima; if not, then the reflector voltage is incorrect. It is best to adjust both controls simultaneously. When properly adjusted for maximum power, the detector produces one volt

~~We have found the second reflector voltage peak (and the third or fourth cavity size peaks) to be strongest.~~ In the experiment the modulation is left off.

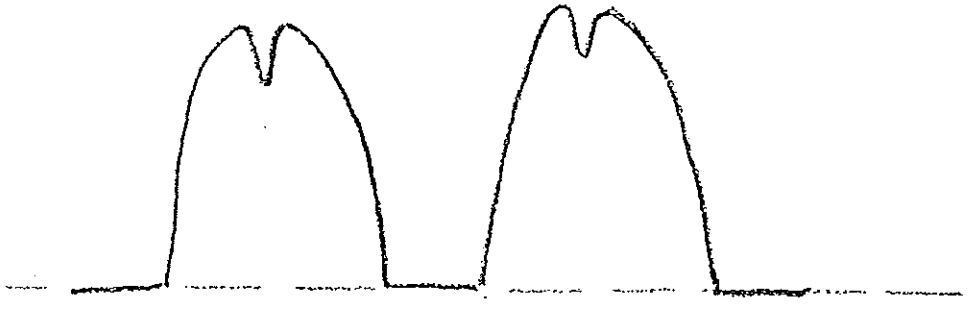


Use of Wave-meter

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The wave-meter is a cavity placed in front of the reflector. When tuned to the proper wavelength it will absorb power and thus produce a pip on the trace as the modulated reflector voltage sweeps past the cavity frequency. When the pip is in the center of the hump (corresponding to the maximum frequency power) the wave meter cavity is in tune. Since in normal operation the modulation is not employed, it should be turned off once the pip is found and the cavity readjusted (the whole line will move.

The present klystron is a K-band oscillator, approximately 24 gigacycles, but the strongest peak is around 22.2 gigacycles. For our data we borrowed a direct-reading model, but it is too expensive for this lab. Another one must be obtained before this experiment may be conducted.



The simplified circuit shows that the voltage across the junction is measured directly by the horizontal amplifier while the current is taken as the voltage induced across a 2.5 resistor in series and displayed on the vertical scale. Where the D.C. resistance of the junction 2.5 and the amplifier equal the trace would appear as a 45 degree line . if contact is not made the trace will be a horizontal line (open-circuit voltage).

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Cooling the Probe

The liquid He dewar may be found on a dolly outside Room 103 LeConte (old section). The rollers should be removed before it is positioned in line with the wave-guide. Before use the entrance must be freed of ice with the hollow aluminum tube that is left on the table.

The probe has an O-ring seal to prevent air from entering the dewar shaft, freezing, and locking the probe in place. If this O-ring seal is held next to the dewar opening, the probe may be <u>slowly</u> pushed into the helium with little waste. If done properly the helium vapor emerges only from the one-way slit in the surgical rubber tubing. It is best to have a TA on hand.

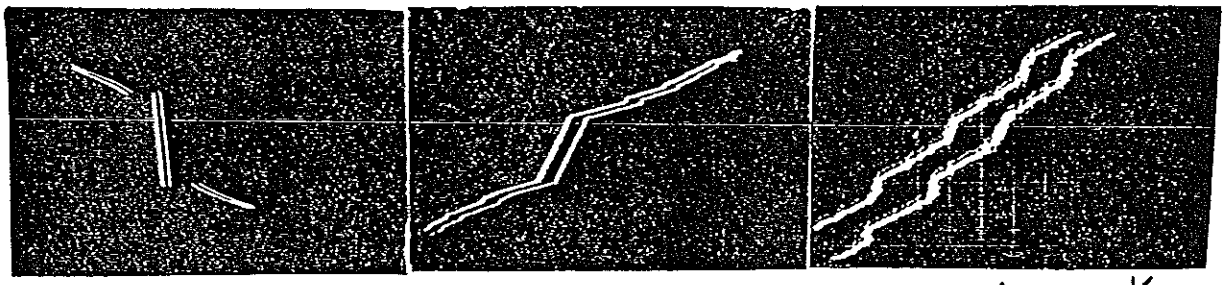
Before the dewar is returned, the volume must be measured. The thin metal tube attached to the dewar has a rubber membrane which vibrates at different frequencies above or below the level of the helium (above--high amplitude 100 cps., below--low amplitude 30 cps.). The graph in the log book converts the measured

distance from ice pack to helium level into liters present in the dewar. With care, waste may be kept below one-half liter; more than 2 liters indicates poor technique.

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Experimental Procedure

When a good Josephson junction has been found (and microwave power applied) the V-I plot reveals a uniform pattern of steps along the circuit load line, the difference between each step being $V = \frac{h\nu}{2e}$.



(reversed) #1 #2 $V = h\nu/2e$ #3

A good junction may be difficult to obtain and it is advisable to have a number of points available. To find the junction the center rod is slowly lowered until there is a jump in the scope trace. The pressure is kept at minimum because the oxide layer may be easily punctured. Once contact is made, the rod is gently twisted. When trace number one (above) is found, microwave power is applied ("100%" on the attenuator means full transmission). Trace number two should appear. If not, there is probably a superconducting short (i.e. the central portion will flatten out due to the critical field applied by the microwaves). The small impedance matching plunger may be pushed in to better couple the microwaves to the junction. Best results occur when the plunger almost

shorts against the central rod (if it shorts the trace becomes a vertical line). It may take some searching to find a suitable junction. Once a trace similar to number two appears, adjustment of the attenuator for maximum transmission and of the plunger for best coupling should produce the desired steps.

For measurement of the voltage between steps the scope is adjusted to include the maximum number of steps in the 10 cm. width and then calibrated, using the "calibrated output." A cable has been constructed to go from the calibrated output to the input of the external horizontal amplifier for this purpose.

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DATA, ERROR ANALYSIS AND FURTHER SUGGESTIONS

By

E. I. Caustin, Jr.

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ERROR ANALYSIS

DATA

N	# OF STEPS	CENTIMETERS	$\frac{CM}{STEP} = X$
1	28.5	7	.2456
2	27.5	7	.2545
3	33	8	~~.2424~~
4	31	7	.2484
5	32	7.6	.2375
6	33	8	.2424
7	23	6	.2609
8	28	7	.2500
9	29	7	.2414
10	29	7	.2414
11	20	5	.2500
12	32.5	8	.2462
13	11.5	3	.2609
14	14.	4	.2857 *

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$\Sigma = 3.2216$

$n = 13$ *

$$\bar{X} = \frac{3.2216}{13} = .2478$$

* THE FOURTEENTH DATA POINT IS DISCARDED BECAUSE IT IS APPROXIMATELY FIVE STANDARD DEVIATIONS [illegible] THE AVERAGE

COMPUTATION OF STANDARD DEVIATION

N	$(\delta = x - \bar{x}) \times 10^4$	$\delta^2 \times 10^8$
1	- 22	484
2	+ 67	4489
3	- 54	2916
4	+ 6	36
5	-103	10609
6	- 54	2916
7	+131	17161
8	+ 22	484
9	- 64	4096
10	- 64	4096
11	+ 22	484
12	- 16	256
	+131	
13		17161

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$$\sum \delta^2 = 65188 \times 10^{-8}$$

$$\sigma = \sqrt{\frac{\sum \delta^2}{n-1}} = (6.52)^{\frac{1}{2}} (.289) \times 10^{-2}$$

$$= 7.4 \times 10^{-3}$$

$$X = \frac{cm}{STEP} = .2478 \pm .0074 \frac{cm}{STEP}$$

CALIBRATION OF OSCILLOSCOPE SCALE

$$\frac{2\ mv}{9.05\ cm} = .2210\ \frac{mv}{cm} = 2.210 \times 10^{-4}\ \frac{volt}{cm}$$

COMPUTATION OF VOLTS/STEP = ΔV

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$$\Delta V = .2478\ \frac{cm}{STEP} \times 2.210 \times 10^{-4}\ \frac{VOLTS}{cm} = 5.476 \times 10^{-5}\ \frac{VOLTS}{STEP}$$

FREQUENCY MEASUREMENT

$$\nu = 22.28 \times 10^{9}\ cps \quad \text{(MEASURED THREE TIMES)}$$

How?

COMPUTATION OF $\frac{\nu}{\Delta V}$

$$\frac{22.28 \times 10^{9}}{5.476 \times 10^{-5}} = 4.069 \times 10^{14}\ \frac{CPS}{VOLT}$$

Error?

COMPUTATION OF THEORETICAL VALUE

$$\frac{2e}{h \cdot 300}\ * = \frac{2 \times 4.803 \times 10^{-10}\ esu}{6.625 \times 10^{-27}\ erg\text{-}sec \quad 3 \times 10^{2}}$$

$$= 4.833 \times 10^{14}\ \frac{CPS}{VOLT}$$

* THE 300 IS A FACTOR USED TO CHANGE FROM GAUSSIAN TO PRACTICAL UNITS

COMPARISON OF THEORETICAL AND EXPERIMENTAL VALUES

$$\begin{array}{r} 4.833 \\ 4.069 \\ \hline .764 \end{array} \qquad \frac{.764}{4.833} \times 100 = 15.8\% \text{ DISCREPANCY}$$

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For experimental values measured directly see W.H. Parker, B.N. Taylor & D.N. Langenberg

"Measurement of 2e/h Using the ac Josephson Effect ..."

Physical Review Letters 18, 287 (Feb. 20, 1967)

Their value is

$$\frac{2e}{h} = 483.5912 \pm .0030 \quad \frac{\text{Mega Hertz}}{\text{micro-volt}}$$

(one Mega Hertz = one kilo mega cycle = 10^9 cps)

ERROR ANALYSIS

see comments on last page for what I consider source of error. -gs

The discrepancy between the experimental and the theoretical value of $\frac{e}{h}$ is too high to be attributed to the measuring instruments or their calibration. These latter errors would be approximately: 1% at most for the wavemeter, 1-5% error for the calibrated voltage output on the oscilloscope and the measurement of that instument's horizontal gain. These together would ~~seem~~ to contribut approximately 2-5% error, not nearly the discrepancy we are faced with. It should be pointed out that the completion of several more runs of the experiment would reduce these errors by means of their statistical nature. However, we were unable to complete more than a single run of the quantitative experiment.

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The day after this single run was made, one of the experimenters found that a wire in the probe structure had been broken, and that the top and bottom segments of this wire were uninsulated and were ~~in~~ twisted. It could be inferred that this wire was broken during the experimental measurement by the fact that it was very difficult to achieve the a-c Josephson pattern on the screen. At the time of the measurements, however, this difficulty was attributed to the difficulty of finding the proper spot for the probe tip on the flattened niobium wire. Undoubtedly, more experience with the junction would have enabled us to more quickly

suspect the broken wire as the trouble.

The pertinence of this broken wire is that if it were to have come in contact with the stainless steel tube when the experiment was being carried out, there would have been a small resistance of the current circuit placed in series with the voltage leads. (See diagram #1.) This would have the effect of causing a slight slope in the zero resistance vertical portions of the Josephson step curve. A difference in the slope from vertical of 10-15% would not have been immediately visible due to the small absolute height of the steps. Unfortunately for the two reasons the slope was not measured on a higher scale of oscilloscope amplification. This omission was due to two reasons. Firstly, when the Josephson pattern with its apparently vertical, zero resistance step, had been produced on the oscilloscope screen, it was wrongly assumed that the probe was operating properly. Secondly, and most frustrating was, that we ran out of time to complete and analyze all of the data we were hoping to. It may be suggested that future workers on this experiment, after obtaining the Josephson step curve, should immediately check the slope of the vertical step segment on a high gain of the scope in order to spot the possibility of a shorted center wire which would be indicated by a non-vertical step. What makes this interpretation of the discrepancy even more reasonable is the fact that when Dr. Richards found that the wire had broken once before, he "make-shifted" a jumper which connected the voltage

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lead to the current source output electrically before the latter was even connected to the probe. This had the effect of introducing a severe change in the slope of the zero resistance step segment.

SUGGESTIONS

1. The center wire can be replaced with some sort of flexible and better insulated (present wire uses brittle lacquer insulation) center wire. Dr. Richards suggested one type of flexible wire; however, it is uninsulated and this might present a problem. There should be some kind of wire to fill these conditions, however it might be necessary to purchase it specially.

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2. Supporting blocks for the waveguide which goes from the klystron oscillator to the probe should be built in order to avoid the necessity of "make-shifting" a support with the inherent dangers of letting the waveguide slip and thus damaging the klystron.

3. A good wavemeter of the type found in Jeffries' and Kittel's group should be gotten especially for use in this experiment. The home-built waveguide we were originally furnished with did not cover the range in which we had to operate. Our measurements were taken at 22.28 Gc and the range of the klystron is about 21-25 Gc. Since frequency measurement accuracy is of high importance, it should be a reliable wavemeter.

4. It should be pointed out that on the older of the two Tektronix Preamplifiers (the one without the on-off panel light) the D-C level control does not function, and the amplification is approximately 4000. According to Dr. Richards these amplifiers have been modified in order to cut down the signal-to-noise ratio. One of the modifications could have eliminated the D-C level control, but the cause of four- fold again is unknown.

5. It is necessary to recognize at the start that both preamplifiers must be operated with the highest possible bandwidth (highest high-frequency cut-off and lowest low frequency cut-off). This is due to the high bandwidth of Fourier components necessary to construct the vertical to horizontal step structure of the Josephson curve.

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6. The oscilloscope needs to be repaired. a) The left channel does not seem to operate at all. b) The triggering controls do not appear to function on any of the triggering selection positions. c) The horizontal sweep is intermittant.

7. A brief suggestion for an interesting possible experiment with the double Josephson junction can be made. A plug for the bottom of the probe could be constructed with two flattened pieces of niobium wire, and a moveable plug could be fitted with two niobium wire points and a set of wires to create a magnetic field. (See diagram #2.) In this way there would be a finite measurable area between two Josephson junctions, and with use of the variable magnetic fields from the fixed wires, the effect of flux

antization should be more easily measurable. The most difficult part would be the geometry of the magnetic field wires (made difficult by the constriction of getting the probe through the narrow neck of the liquid helium dewer neck) and the computation of the magnetic flux through the effective area of the double-junction. However, these problems with some faculty help would not be unsurmountable difficulties.

5. There was a double trace on the oscilloscope of the Josephson curve. This can be accounted for by the phase shift caused by the leads' and probe's stray inductances and capacitances. Although this double trace does not hinder measurement appreciably, esthetically it would be nicer to have a single trace. This could be done by placing the appropriate inductance or capacitance in the circuit to create another intentional phase shift in the opposite direction of the inherent one. Probably the experimental use of a capacitance and inductance substitution box or their equivalents would quickly lead to the appropriate values and placement of this phase shifter.

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Not true. This ac pickup is a serious source of systematic error. It adds to the true voltage across the junction an additional voltage which is not constant, not even linear, with time, and with arbitrary phase shifts. This means that the spacing between Josephson jumps is different for upper and lower traces of the loop -- different even between different jumps on the same trace.

1) Efforts should be made to eliminate this pickup.
2) Averages of the jump spacing must be made over the